On the Structure of Classical Dynamics

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Abstract

The quadratic curvature density of compatibility geometry determines a classical phase-space realization and a covariant presymplectic current whose hypersurface pullback agrees, after Legendre substitution, with the canonical split symplectic form.

The induced classical system carries canonical variables, a Hamiltonian, the internal momentum map, the Gauss constraint, and a formal reduced Poisson algebra. It is closed under internal gauge Hamiltonian evolution, propagates the Gauss surface, and yields a projected linearized principal symbol together with a mixed-sector kernel at flat compatible $K^{(0)} = 0$ backgrounds, a reduced phase-space count, and a quadratic Hamiltonian sign classification.

In codimension one, the contracted Gauss and Codazzi identities reduce, in the fixed-metric internal-gauge regime, to the standard hypersurface constraint equations.

Introduction

An admissible orthogonal compatibility connection on

$$E = TM \oplus N$$

with frozen Levi–Civita tangent block and dynamical normal and mixed sectors is the field object. Its dynamical variables are the normal connection component

$$A \in \Omega^1(M, \mathfrak{so}(N))$$

and the mixed compatibility field

$$K \in \Gamma(\mathrm{Sym}^2 T^*M \otimes N),$$

which determines the off-diagonal admissible block [1]. The corresponding quadratic curvature density gives the Euler–Lagrange equation [2]

$$D^{\oplus} \star \mathcal{F}^{\oplus} = 0.$$

The classical phase-space realization of this law determines the structures developed below.

The formal Hamiltonian realization of the fixed covariant law consists of the variational current, the presymplectic current, canonical variables, the momentum map, the constraint surface, Hamiltonian flow, and the reduced Poisson algebra. Closure, projected symbol structure, degree count, and quadratic sign classification are determined at formal Hamiltonian level and, for the linearized and quadratic analyses, about the admissible backgrounds specified below.

Assumption 0.1 (Standing Regime)**.** The analysis is carried out under the following standing regime.

(i) The tangent metric h and the normal metric g_N are fixed background data.

(ii) The tangent block of the admissible connection is the Levi–Civita connection of h.

(iii) Only the internal normal gauge group $\mathcal{G}_N = \Gamma(\mathrm{SO}(N))$ is quotiented. Diffeomorphisms are not quotiented.

(iv) A fixed product split is assumed whenever canonical formulas are used.

(v) Either Σ is compact without boundary or all fields, variations, and gauge parameters satisfy a decay regime eliminating boundary terms from the integrations by parts used below.

(vi) All phase-space constructions are treated formally on the space of smooth admissible fields and variations.

(vii) No Banach-, Fréchet-, or global quotient-manifold structure is assumed.

Notation. Write $\mathcal{A}_\oplus$ for the admissible affine configuration space of orthogonal compatibility connections, $\mathcal{G}_N$ for the internal normal gauge group, $\mathcal{F}^\oplus$ for the curvature of the admissible connection, and $D^\oplus$ for the corresponding covariant exterior derivative. The canonical invariant pairing on $\mathfrak{so}(E)$ is

$$\langle X, Y\rangle_{\mathrm{can}} := -\mathrm{tr}(XY).$$

For Lie-algebra-valued differential forms, $\langle\cdot,\cdot\rangle_{\mathrm{can}}$ denotes the induced fiberwise pairing composed with the Hodge contraction defining the density L_{curv}. On split variables, $\langle\cdot,\cdot\rangle_{\mathrm{can}}$ denotes the induced canonical pairing. Write

$$L_{\mathrm{curv}} = \frac{1}{2}\left\langle \mathcal{F}^\oplus, \star\mathcal{F}^\oplus\right\rangle_{\mathrm{can}}.$$

$(D^\oplus)^\dagger$ denotes the formal adjoint of $D^\oplus$ with respect to this Hodge-paired density convention. Relative to the orthogonal splitting $E = TM \oplus N$, the curvature decomposes as

$$\mathcal{F}^\oplus = \begin{pmatrix} \mathcal{G} & -\mathcal{C}^* \\ \mathcal{C} & \mathcal{S} \end{pmatrix},$$

where $\mathcal{G}$, $\mathcal{C}$, and $\mathcal{S}$ are the Gauss, Codazzi, and Ricci curvature channels.

Proposition 0.2 (Cited Inputs)**.** *The following inputs from [1, 2] are used.*

(1) The admissible orthogonal compatibility connection, its affine configuration space $\mathcal{A}_\oplus$, its internal gauge symmetry $\mathcal{G}_N$, and its curvature block decomposition are established in [1].

(2) The quadratic curvature density

$$L_{\mathrm{curv}} = \frac{1}{2}\left\langle \mathcal{F}^\oplus, \star\mathcal{F}^\oplus\right\rangle_{\mathrm{can}}$$

and the covariant Euler–Lagrange equation

$$D^\oplus \star \mathcal{F}^\oplus = 0$$

are established in [2].

(3) The associated Noether identity

$$(D^\oplus)^\dagger \mathcal{E} = 0$$

for the Euler operator $\mathcal{E}$ is established in [2].

1 Covariant Phase Geometry

1.1 Dynamical Field Equation

The dynamical field is the admissible orthogonal compatibility connection

$$\nabla^{\oplus} = \nabla^{h} \oplus \nabla^{\perp} + \Psi(K)$$

on $E = TM \oplus N$ with curvature

$$\mathcal{F}^{\oplus} = \begin{pmatrix} \mathcal{G} & -\mathcal{C}^{*} \\ \mathcal{C} & \mathcal{S} \end{pmatrix}.$$

The field equation fixed in [2] is

$$D^{\oplus} \star \mathcal{F}^{\oplus} = 0.$$

With respect to the fixed density convention, the Euler equation takes the adjoint form

$$(D^{\oplus})^{\dagger} \mathcal{F}^{\oplus} = 0.$$

1.2 Classical Configuration Space

Definition 1.1 (Classical Configuration Space)**.** The classical configuration space is the admissible affine space

$$\mathcal{A}_{\oplus}$$

of orthogonal compatibility connections with frozen tangent block.

An admissible tangent vector at $\omega^{\oplus} \in \mathcal{A}_{\oplus}$ has block form

$$\delta\omega^{\oplus} = \begin{pmatrix} 0 & -\delta K^{*} \\ \delta K & \delta A \end{pmatrix},$$

where $\delta A \in \Omega^{1}(M, \mathfrak{so}(N))$ and $\delta K \in \Gamma(\mathrm{Sym}^{2} T^{*}M \otimes N)$.

1.3 Variational Current

Proposition 1.2 (Covariant Variational Identity)**.** *Under Assumption 0.1, the first variation of the quadratic curvature density is*

$$\delta L_{\mathrm{curv}} = \left\langle \delta\omega^{\oplus}, (D^{\oplus})^{\dagger} \mathcal{F}^{\oplus}, + \right\rangle_{\mathrm{can}} \mathrm{d}\Theta_{L}(\delta\omega^{\oplus}),$$

with covariant variational current

$$\Theta_{L}(\delta\omega^{\oplus}) = \left\langle \delta\omega^{\oplus}, \star\mathcal{F}^{\oplus} \right\rangle_{\mathrm{can}}.$$

Proof. Since $\delta\mathcal{F}^{\oplus} = D^{\oplus}(\delta\omega^{\oplus})$, the first variation is $\delta L_{\mathrm{curv}} = \langle D^{\oplus}(\delta\omega^{\oplus}), \star\mathcal{F}^{\oplus} \rangle_{\mathrm{can}}$, and formal integration by parts with respect to the Hodge-paired density gives the stated identity. □

Corollary 1.3 (Admissible Block Form of the Variational Current)**.** *With the admissible block decomposition,*

$$\Theta_{L}(\delta\omega^{\oplus}) = -\langle \delta A, \star\mathcal{S} \rangle_{\mathrm{can}} + 2 \langle \delta K, \star\mathcal{C} \rangle_{\mathrm{can}}.$$

Proof. Insert the admissible variation block and the curvature block into the pairing of Proposition 1.2. The two off-diagonal mixed entries contribute equally under the canonical trace pairing, hence the factor 2. □

2 Symplectic Structure

2.1 Presymplectic Current

Definition 2.1 (Covariant Presymplectic Current)**.** For admissible variations $\delta_1\omega^{\oplus}, \delta_2\omega^{\oplus}$, define

$$\omega_L(\delta_1, \delta_2) := \delta_1\Theta_L(\delta_2) - \delta_2\Theta_L(\delta_1).$$

Proposition 2.2 (Admissible Covariant Presymplectic Current)**.** *Under Assumption 0.1,*

$$\omega_L(\delta_1, \delta_2) = -\delta_1 \langle \delta_2 A, \star\mathcal{S}\rangle_{\text{can}} + \delta_2 \langle \delta_1 A, \star\mathcal{S}\rangle_{\text{can}} + 2\delta_1 \langle \delta_2 K, \star\mathcal{C}\rangle_{\text{can}} - 2\delta_2 \langle \delta_1 K, \star\mathcal{C}\rangle_{\text{can}}.$$

Proof. Antisymmetrize the current of Corollary 1.3. The derivation is written out in Appendix B. □

2.2 Hypersurface Symplectic Form

Fix the product decomposition of Assumption 0.1(iv). For any spatial slice $\Sigma_t \cong \Sigma$, define

$$\Omega_{\Sigma_t}(\delta_1, \delta_2) := \int_{\Sigma_t} \omega_L(\delta_1, \delta_2).$$

Proposition 2.3 (Formal Hypersurface Presymplectic Form)**.** *Under Assumption 0.1, Ω_{Σ_t} is a formal hypersurface presymplectic form. If $\delta_\epsilon\omega^{\oplus} = D^{\oplus}\epsilon$ is an infinitesimal internal gauge variation and the field equation together with the Noether identity hold, then*

$$\iota_{\delta_\epsilon}\Omega_{\Sigma_t}$$

is an exact boundary term whose slice integral vanishes under Assumption 0.1(v). Consequently internal gauge directions are degeneracy directions of Ω_{Σ_t}.

Proof. The existence of the slice integral is the boundary statement in Assumption 0.1(v). The gauge-degeneracy statement follows from the Noether identity after integrating the total derivative obtained by contracting the presymplectic current with δ_ϵ. Appendix B gives the formal identity. □

Proposition 2.4 (Slice Form of the Covariant Presymplectic Current)**.** *Under the fixed product split, the pullback of the covariant presymplectic current to Σ_t kills the dt-magnetic terms and depends only on the pulled-back electric channels $E_\perp$ and E_K, which are precisely the channels entering the canonical momenta below.*

Proof. Appendix B decomposes the relevant curvature-star blocks as

$$\star\mathcal{S} = *_\Sigma E_\perp + dt \wedge *_\Sigma B_\perp, \qquad \star\mathcal{C} = *_\Sigma E_K + dt \wedge *_\Sigma B_K.$$

Hence the hypersurface pullback is

$$\iota_t^*(\star\mathcal{S}) = *_\Sigma E_\perp, \qquad \iota_t^*(\star\mathcal{C}) = *_\Sigma E_K.$$

Therefore the pullback annihilates the temporal dt-terms and retains only the electric coefficients. □

Under Assumption 0.1, the hypersurface current agrees with the canonical split symplectic form after the Legendre substitution:

$$p_A = E_\perp, \qquad p_K = 2E_K$$

Theorem 2.5 (Covariant-to-Canonical Symplectic Bridge)**.**

$$\iota_t^*\omega_L = \delta E_\perp \wedge \delta A + 2\delta E_K \wedge \delta K.$$

$$\Omega_{\Sigma_t} = \int_{\Sigma_t} \omega_L = \int_\Sigma (\delta p_A \wedge \delta A + \delta p_K \wedge \delta K) = \Omega.$$

Proof. Combine Proposition 2.4 with the slice pullback identity

$$\Omega_{\Sigma_t} = \int_\Sigma \big(\delta E_\perp \wedge \delta A + 2\delta E_K \wedge \delta K\big)$$

from Appendix B. Substituting $p_A = E_\perp$ and $p_K = 2E_K$ yields the displayed canonical form. The detailed computation is written out in Appendix B. □

2.3 Gauge Degeneracies

Remark 2.6 (Formal versus Strong Symplectic Language)**.** The object obtained in Proposition 2.3 is presymplectic in the formal field-theoretic sense fixed by Assumption 0.1(vi).

3 Canonical Decomposition

3.1 Spacetime Foliation

Assumption 3.1 (Split Geometry)**.** All split formulas in Sections 3–5 and Appendices C–J are written in the fixed product background

$$M \cong \mathbb{R} \times \Sigma, \qquad h = dt^2 + \gamma,$$

with unit lapse, zero shift, and orientation chosen so that $\mathrm{vol}_h = dt \wedge \mathrm{vol}_\gamma$. The Hodge operator $\star$ is taken with these conventions, and $*_\Sigma$ denotes the Hodge operator of (Σ, γ). All split identities below are statements only in this fixed product geometry; no foliation-covariant statement is intended.

3.2 Spatial and Temporal Components

Decompose the admissible connection as

$$\omega^\oplus = \omega_0^\oplus\, dt + \omega_\Sigma^\oplus,$$

where

$$\omega_0^\oplus \in \Gamma(\mathfrak{so}(E)), \qquad \omega_\Sigma^\oplus \in \Omega^1(\Sigma, \mathfrak{so}(E)).$$

The temporal block consists of multiplier fields. The spatial block carries the canonical variables after the Legendre transform.

The mixed field is symmetric in its tangent slots at spacetime level,

$$K \in \Gamma(\mathrm{Sym}^2 T^*M \otimes N).$$

In the fixed product split,

$$K = K_{00}\, dt \otimes dt + 2K_{0i}\, dt \odot dx^i + K_{ij}\, dx^i \odot dx^j, \qquad K_{ij} = K_{ji}.$$

Only the spatial symmetric block

$$K_\Sigma := K_{ij}\, dx^i \odot dx^j \in \Gamma(\mathrm{Sym}^2 T^*\Sigma \otimes N)$$

is retained. Only K_Σ enters the canonical system, since it is the only mixed block carrying an independent kinetic term in the split Lagrangian. From this point onward, K denotes K_Σ. The components K_{00} and K_{0i} are nondynamical split data.

3.3 Canonical Variables

Definition 3.2 (Electric and Magnetic Split)**.** The electric and magnetic channels inherit the normal/mixed admissible block structure of $\mathcal{F}^{\oplus}$. Let $E_i^{\oplus} := (\mathcal{F}^{\oplus})_{0i}$ denote the electric curvature channel and $B_{ij}^{\oplus} := (\mathcal{F}^{\oplus})_{ij}$ the magnetic curvature channel. With respect to the admissible block decomposition,

$$E_\perp \in \Omega^1(\Sigma, \mathfrak{so}(N)), \qquad E_K \in \Omega^1(\Sigma, \mathrm{Hom}(T\Sigma, N)),$$

$$B_\perp \in \Omega^2(\Sigma, \mathfrak{so}(N)), \qquad B_K \in \Omega^2(\Sigma, \mathrm{Hom}(T\Sigma, N)), \qquad B_{TT} \in \Omega^2(\Sigma, \mathfrak{so}(T\Sigma)).$$

$$\star\mathcal{F}^{\oplus} = \mathcal{E}_\star^{\oplus} + dt \wedge \mathcal{B}_\star^{\oplus},$$

where $\mathcal{E}_\star^{\oplus} := *_\Sigma \mathcal{E}^{\oplus}$ is the spatial Hodge dual of the electric block and $\mathcal{B}_\star^{\oplus} := *_\Sigma B^{\oplus}$ is the spatial Hodge dual of the magnetic block.

In the symmetric mixed sector, E_K is regarded as an element of $\Gamma(\mathrm{Sym}^2 T^*\Sigma \otimes N)$.

Definition 3.3 (Magnetic Potential Functional)**.** Define the magnetic potential functional

$$V_{\mathrm{mag}} := \frac{1}{2} \int_\Sigma \left\langle B^{\oplus}, B^{\oplus} \right\rangle_{\mathrm{can}} \mathrm{vol}_\gamma.$$

Proposition 3.4 (Split Curvature Lagrangian Functional)**.** *In the fixed product decomposition of Assumption 3.1, the spatially integrated quadratic curvature functional takes the split form*

$$L = \int_\Sigma \frac{1}{2} \left(E_\perp^2 + 2E_K^2 \right) \mathrm{vol}_\gamma - V_{\mathrm{mag}}.$$

Proof. The spatially integrated curvature functional separates into electric and magnetic channels under the chosen foliation. The mixed coefficient 2 comes from the same block-trace multiplicity that already appeared in Corollary 1.3. Appendix C gives the split calculation. □

Proposition 3.5 (Canonical Momenta)**.** *The canonical momenta of the spatial admissible variables are*

$$p_A \in \Omega^1(\Sigma, \mathfrak{so}(N)), \qquad p_K \in \Gamma(\mathrm{Sym}^2 T^*\Sigma \otimes N),$$

and satisfy

$$p_A = \frac{\partial L}{\partial E_\perp} = E_\perp, \qquad p_K = \frac{\partial L}{\partial E_K} = 2E_K.$$

Proof. Differentiation gives the stated momenta. □

Proposition 3.6 (Legendre Map and Formal Canonical Forms)**.** *In the fixed product regime, the Legendre transform determined by Proposition 3.5 is only a formal pointwise split-variable inversion in the variables $(E_\perp, E_K)$, with inverse*

$$E_\perp = p_A, \qquad E_K = \frac{1}{2} p_K.$$

Define the Hamiltonian density

$$\mathcal{H} := \frac{1}{2} p_A^2 + \frac{1}{4} p_K^2 + \frac{1}{2} \left\langle B^{\oplus}, B^{\oplus} \right\rangle_{\mathrm{can}}.$$

The corresponding Hamiltonian functional is

$$H := \int_\Sigma \mathcal{H} \, \mathrm{vol}_\gamma.$$

The formal canonical one-form and symplectic form in the split variables are

$$\Theta = \int_\Sigma (p_A\,\delta A + p_K\,\delta K), \qquad \Omega = -\delta\Theta = \int_\Sigma (\delta p_A \wedge \delta A + \delta p_K \wedge \delta K).$$

These are the formal canonical expressions in the split variables.

Proof. The kinetic block of Proposition 3.4 is diagonal and nondegenerate in the variables $(E_\perp, E_K)$, so the displayed pointwise inversion follows. The remaining formulas follow from the explicit Legendre transform of Proposition 3.4; see Appendix C. □

4 Constraints and Momentum Map

4.1 Gauge Action on Phase Space

Let $\mathfrak{g}_N = \Gamma(\mathfrak{so}(N))$. For $\epsilon \in \mathfrak{g}_N$, the infinitesimal internal gauge action is

$$\delta_\epsilon A = D_A\epsilon, \qquad \delta_\epsilon K = \epsilon \cdot K, \qquad \delta_\epsilon p_A = [\epsilon, p_A], \qquad \delta_\epsilon p_K = \epsilon \cdot p_K.$$

Here D_A denotes the spatial covariant derivative determined by A, and the gauge algebra acts on the N-valued part of the mixed sector by the defining representation.

4.2 Momentum Map

Proposition 4.1 (Momentum Map)**.** *The internal gauge action on the formal canonical phase space is Hamiltonian. Let $\mathfrak{g}_N^\vee$ denote the algebraic dual of $\mathfrak{g}_N = \Gamma(\mathfrak{so}(N))$, paired with $\mathfrak{g}_N$ by the integrated canonical trace form under Assumption 0.1. Define the momentum map*

$$J : T^*\mathcal{A}_\oplus \to \mathfrak{g}_N^\vee$$

defined by

$$\langle J(A, K, p_A, p_K), \epsilon\rangle := \int_\Sigma \langle \epsilon, D_A p_A + p_K K^* - K p_K^*\rangle_{\mathrm{can}}\ \mathrm{vol}_\gamma.$$

For every $\epsilon \in \mathfrak{g}_N$,

$$\mathrm{d}\langle J, \epsilon\rangle = \iota_{X_\epsilon}\Omega.$$

Proof. Differentiate the defining functional and compare it with the contraction of Ω against the fundamental vector field X_ϵ. The calculation is recorded in Appendix D. □

Theorem 4.2 (Gauss Constraint)**.** *The Gauss constraint is $J = 0$, equivalently*

$$\langle J, \epsilon\rangle = 0 \quad \textit{for all } \epsilon \in \mathfrak{g}_N.$$

and hence

$$D_A p_A + p_K K^* - K p_K^* = 0.$$

Proof. The functional J vanishes if and only if its integrated canonical pairing with every $\epsilon \in \mathfrak{g}_N$ vanishes. Under the algebraic dual pairing fixed in Proposition 4.1, vanishing of the integrated pairing against every $\epsilon \in \mathfrak{g}_N$ is equivalent to vanishing of the displayed Gauss expression. □

4.3 Constraint Surface

Definition 4.3 (Constraint Surface)**.** The Gauss constraint surface is

$$\mathcal{S}_{\mathrm{G}} := J^{-1}(0).$$

$\mathcal{S}_{\mathrm{G}}$ is treated as a formal zero locus together with its linearized tangent equation. No global coisotropic theorem is asserted.

5 Hamiltonian Dynamics

5.1 Curvature Hamiltonian

The Legendre transform of the split Lagrangian gives the Hamiltonian density

$$\mathcal{H} = \frac{1}{2}p_A^2 + \frac{1}{4}p_K^2 + \frac{1}{2}\left\langle B^{\oplus}, B^{\oplus}\right\rangle_{\mathrm{can}},$$

and the Hamiltonian functional is

$$H = \int_{\Sigma} \mathcal{H}\,\mathrm{vol}_{\gamma}.$$

5.2 Hamiltonian Flow

Theorem 5.1 (Hamiltonian Evolution)**.** *The Hamilton equations of H are*

$$\dot{A} = \frac{\delta H}{\delta p_A} = p_A, \qquad \dot{K} = \frac{\delta H}{\delta p_K} = \frac{1}{2}p_K,$$

$$\dot{p}_A = -\frac{\delta H}{\delta A}, \qquad \dot{p}_K = -\frac{\delta H}{\delta K}.$$

Eliminating p_A and p_K yields the second-order Euler system determined by the split Lagrangian functional of Proposition 3.4.

Proof. The canonical bracket gives the first-order system, and the Legendre relations $p_A = E_{\perp}$, $p_K = 2E_K$ convert it to the Euler system of Proposition 3.4. □

5.3 Projected Decomposition of $D^{\oplus} \star \mathcal{F}^{\oplus} = 0$

Assumption 3.1 is in force. Decompose $D^{\oplus} \star \mathcal{F}^{\oplus}$ into temporal and spatial channels.

Proposition 5.2 (Projected Foliated Decomposition of the Covariant Equation)**.** *Under Assumptions 0.1 and 3.1, the temporal and spatial projections of the equation*

$$D^{\oplus} \star \mathcal{F}^{\oplus} = 0$$

in the fixed product split are:

$$D_{\omega_\Sigma}\mathcal{E}_{\star}^{\oplus} = 0,$$

$$\partial_t \mathcal{E}_{\star}^{\oplus} + [\omega_0^{\oplus}, \mathcal{E}_{\star}^{\oplus}] - D_{\omega_\Sigma}\mathcal{B}_{\star}^{\oplus} = 0.$$

Proof. Appendix E writes $\star\mathcal{F}^{\oplus}$ in electric and magnetic channels and expands $D^{\oplus} \star \mathcal{F}^{\oplus}$ in the fixed product split. The displayed temporal and spatial projections are exactly the resulting block equations. □

Proposition 5.3 (Temporal Projection and the Gauss Constraint)**.** *Under Assumptions 0.1 and 3.1, the normal projection of the temporal channel*

$$D_{\omega_\Sigma}\mathcal{E}_\star^\oplus = 0$$

of Proposition 5.2 agrees with the Gauss constraint of Theorem 4.2 after the Legendre substitution

$$p_A = E_\perp, \qquad p_K = 2E_K.$$

Proof. The normal block of

$$D_{\omega_\Sigma}\mathcal{E}_\star^\oplus = 0$$

is exactly the divergence-type Gauss equation after substituting the canonical momenta $p_A = E_\perp$ and $p_K = 2E_K$. Appendix E gives the split block computation. □

Theorem 5.4 (Hamiltonian Identification of the Projected Evolution Equations)**.** *Under Assumptions 0.1 and 3.1, at formal algebraic level and on the projected admissible normal and mixed evolution equations, the Hamiltonian system of Theorem 5.1 agrees with the evolution channel of Proposition 5.2 after the Legendre substitution*

$$p_A = E_\perp, \qquad p_K = 2E_K.$$

Proof. Appendix E identifies the normal and mixed admissible projections of the evolution channel

$$\partial_t\mathcal{E}_\star^\oplus + [\omega_0^\oplus, \mathcal{E}_\star^\oplus] - D_{\omega_\Sigma}\mathcal{B}_\star^\oplus = 0$$

with the Hamilton equations after substituting $p_A = E_\perp$ and $p_K = 2E_K$. Proposition 5.3 records the separate temporal-channel identification. This is the claimed projected Hamiltonian identification. □

5.4 Constraint Algebra

Proposition 5.5 (Closure of the Gauss Generators)**.** *For $\epsilon_1, \epsilon_2 \in \mathfrak{g}_N$,*

$$\{\langle J, \epsilon_1\rangle, \langle J, \epsilon_2\rangle\} = \langle J, [\epsilon_1, \epsilon_2]\rangle.$$

Moreover,

$$\{H, \langle J, \epsilon\rangle\} = 0 \qquad \textit{for all } \epsilon \in \mathfrak{g}_N.$$

Proof. The Hamiltonian vector field of $\langle J, \epsilon\rangle$ is the infinitesimal internal gauge action. Hence

$$\{\langle J, \epsilon_1\rangle, \langle J, \epsilon_2\rangle\} = X_{\epsilon_2}\big(\langle J, \epsilon_1\rangle\big) = \langle J, [\epsilon_1, \epsilon_2]\rangle.$$

The second identity is exactly the gauge invariance of H from Proposition 6.3. Appendix E gives the same computation in split variables. □

6 Poisson Structure and Observable Algebra

6.1 Poisson Bracket

Definition 6.1 (Canonical Poisson Bracket)**.** For admissible functionals F, G, define

$$\{F, G\} = \int_\Sigma \left(\frac{\delta F}{\delta A}\frac{\delta G}{\delta p_A} - \frac{\delta F}{\delta p_A}\frac{\delta G}{\delta A} + \frac{\delta F}{\delta K}\frac{\delta G}{\delta p_K} - \frac{\delta F}{\delta p_K}\frac{\delta G}{\delta K} \right).$$

The display is shorthand for the integrated functional Poisson bracket, with $\langle\cdot,\cdot\rangle_{\text{can}}$ suppressed in the integrand and each integrand term paired by the canonical bundle pairing on the corresponding split variable.

6.2 Gauge-Invariant Observables

Definition 6.2 (Gauge-Invariant Observable)**.** An observable F is internally gauge-invariant if

$$\{F, \langle J, \epsilon \rangle\} = 0 \qquad \text{for all } \epsilon \in \mathfrak{g}_N.$$

Proposition 6.3 (Gauge Invariance of the Hamiltonian)**.** *The Hamiltonian H is gauge invariant:*

$$\{H, \langle J, \epsilon \rangle\} = 0 \qquad \textit{for all } \epsilon \in \mathfrak{g}_N.$$

Hence H defines an element of the reduced observable algebra below.

Proof. This is the second identity of Proposition 5.5. □

6.3 Reduced Observable Algebra

Proposition 6.4 (Formal Reduced Observable Algebra)**.** *At the formal algebraic level fixed by Assumption 0.1, the gauge-invariant observables are closed under the canonical Poisson bracket. Their restriction to $\mathcal{S}_{\mathrm{G}}$ inherits a Poisson bracket. This induced bracket is the formal reduced observable algebra.*

Proof.

$$\{\{F, G\}, \langle J, \epsilon \rangle\} = \{\{F, \langle J, \epsilon \rangle\}, G\} + \{F, \{G, \langle J, \epsilon \rangle\}\}.$$

Closure follows from the Jacobi identity together with the defining vanishing brackets against all Gauss generators. Their restriction to $\mathcal{S}_{\mathrm{G}}$ therefore inherits a Poisson bracket. The reduction is formal and algebraic. □

7 Formal Closure and Constraint Propagation

7.1 Differential Closure of the Classical System

Proposition 7.1 (Hamiltonian Closure of the Classical System)**.** *Under Assumptions 0.1 and 3.1, the formal classical system closes in the internal-gauge Hamiltonian sense:*

$$\{\langle J, \epsilon_1 \rangle, \langle J, \epsilon_2 \rangle\} = \langle J, [\epsilon_1, \epsilon_2] \rangle, \qquad \{H, \langle J, \epsilon \rangle\} = 0$$

for all $\epsilon, \epsilon_1, \epsilon_2 \in \mathfrak{g}_N$.

Proof. The first identity is Proposition 5.5. The second is Proposition 6.3. These are the internal-gauge Hamiltonian closure identities. □

7.2 Constraint Propagation

Theorem 7.2 (Propagation of the Gauss Constraint)**.** *Let $(A(t), K(t), p_A(t), p_K(t))$ be a Hamiltonian trajectory of H in the fixed split regime. If the initial data lie on the Gauss surface $\mathcal{S}_{\mathrm{G}}$, then the trajectory remains in $\mathcal{S}_{\mathrm{G}}$ throughout any interval of formal Hamiltonian evolution.*

Proof. For every $\epsilon \in \mathfrak{g}_N$,

$$\frac{d}{dt}\langle J, \epsilon \rangle = \{\langle J, \epsilon \rangle, H\} = 0$$

by Proposition 7.1. Hence every Gauss functional is constant along Hamiltonian flow. If all Gauss functionals vanish initially, they vanish throughout the interval of formal Hamiltonian evolution. □

7.3 Compatibility of Covariant and Canonical Closure

Proposition 7.3 (Compatibility of Covariant and Canonical Closure)**.** *Under Assumptions 0.1 and 3.1, the canonical closure statements of Proposition 7.1 agree with the projected covariant identity structure determined by the equation $D^{\oplus} \star \mathcal{F}^{\oplus} = 0$ and the Noether identity $(D^{\oplus})^{\dagger}\mathcal{E} = 0$. In particular, the temporal projection of the covariant system yields the propagated Gauss channel, while the spatial projected evolution blocks yield the Hamiltonian evolution channel.*

Proof. Proposition 5.3 identifies the temporal projection of the covariant law with the Gauss channel after the Legendre substitution. Theorem 5.4 identifies the projected evolution channel with the Hamiltonian evolution equations. The same split decomposition therefore carries both the preserved Gauss channel and the projected evolution channel. □

8 Propagation and Degrees of Freedom

8.1 Linearization about Admissible Backgrounds

Fix an admissible background $(A^{(0)}, K^{(0)}, p_A^{(0)}, p_K^{(0)})$ satisfying the classical compatibility dynamics in the standing regime. Write perturbations

$$A = A^{(0)} + a, \qquad K = K^{(0)} + \kappa, \qquad p_A = p_A^{(0)} + \pi_A, \qquad p_K = p_K^{(0)} + \pi_K.$$

Proposition 8.1 (Linearized Projected Classical System)**.** *Under Assumptions 0.1 and 3.1, linearization of the projected classical system about an admissible background yields the split first-order system*

$$\delta\big(D_{\omega_\Sigma}\mathcal{E}_\star^{\oplus}\big) = 0 \qquad \textit{(linearized temporal constraint channel)}$$

and

$$\delta\big(\partial_t\mathcal{E}_\star^{\oplus} + [\omega_0^{\oplus}, \mathcal{E}_\star^{\oplus}] - D_{\omega_\Sigma}\mathcal{B}_\star^{\oplus}\big) = 0 \qquad \textit{(linearized projected evolution channel).}$$

Proof. Linearize the projected equations of Proposition 5.2 about the stated background. The two resulting channels are recorded in Appendix G. □

8.2 Principal Symbol and Propagation Type

Proposition 8.2 (Projected Principal Symbol and Special-Background Degeneracy)**.** *For admissible backgrounds in the fixed split regime, the projected linearized evolution system of Proposition 8.1 has principal symbol $\sigma_{\mathrm{proj}}(\tau, \xi)$ acting on the projected normal block a_{proj} and mixed block κ_{proj}.*

(i) At generic admissible backgrounds, $\sigma_{\mathrm{proj}}(\tau, \xi)$ is quadratic in the dual split covector variables (τ, ξ) on the projected normal block a_{proj} and mixed block κ_{proj}.

(ii) At flat compatible backgrounds with $K^{(0)} = 0$, the mixed projected block retains the reduced-symbol degeneracy identified in [1]; in particular, the projected mixed-block kernel direction $\kappa_{\mathrm{deg}}(\xi)$ satisfies

$$\sigma_{\mathrm{proj}}(\tau, \xi)\begin{pmatrix} 0 \\ \kappa_{\mathrm{deg}}(\xi) \end{pmatrix} = 0.$$

The field $\kappa_{\mathrm{deg}}(\xi)$ is therefore a mixed-block kernel direction of the projected principal symbol.

Proof. Appendix G records the projected principal symbol obtained by replacing the linearized split derivatives with the dual covector variables (τ, ξ). The generic projected symbol is quadratic in (τ, ξ) on the projected normal block a_{proj} and mixed block κ_{proj}. The sharpened flat-background degeneracy follows from the special-background principal-symbol analysis already established in [1]. The field $\kappa_{\mathrm{deg}}(\xi)$ is a mixed-block kernel direction of the projected principal symbol. □

8.3 Degree-of-Freedom Count

Proposition 8.3 (Reduced Classical Phase-Space Count)**.** *Let $n = \dim \Sigma$ and $k = \mathrm{rank}(N)$. Then the spatial normal-connection variables contribute*

$$n\frac{k(k-1)}{2}$$

configuration components per point, while the spatial mixed variables contribute

$$\frac{n(n+1)}{2}k$$

configuration components per point. Hence the unreduced canonical phase space has

$$2\left(n\frac{k(k-1)}{2} + \frac{n(n+1)}{2}k\right)$$

components per point. Subtracting the $2\dim \mathfrak{so}(N) = k(k-1)$ phase-space components removed by the first-class Gauss generators yields

$$2\left(n\frac{k(k-1)}{2} + \frac{n(n+1)}{2}k\right) - k(k-1)$$

reduced classical phase-space components per spatial point. This is the formal reduced classical phase-space count per spatial point in the fixed-metric internal-gauge theory; no diffeomorphism subtraction is included.

Proof. The count follows by direct enumeration of the canonical phase-space variables A_Σ and K_Σ, doubling to pass to canonical phase space, and subtracting twice the dimension of the internal gauge algebra because the Gauss generators are first class. Appendix H records the full count. □

9 Quadratic Sign Classification

This section gives the quadratic Hamiltonian sign classification in the fixed split regime.

9.1 Quadratic Hamiltonian

Definition 9.1 (Quadratic Hamiltonian about an Admissible Background)**.** Fix an admissible background in the standing regime. The quadratic Hamiltonian is the second-order part of the split Hamiltonian in the perturbation variables $(a, \kappa, \pi_A, \pi_K)$, with background-dependent terms collected into $H^{(2)}_{\mathrm{bg}}$. In the fixed split regime it has the formal quadratic split form

$$\begin{aligned} H^{(2)} = \frac{1}{2}\|\pi_A\|^2 + \frac{1}{4}\|\pi_K\|^2 + \frac{1}{2}\|B^{(1)}\|^2 \\ + H^{(2)}_{\mathrm{bg}}. \end{aligned}$$

9.2 Positivity and Indefinite Channels

Proposition 9.2 (Quadratic Hamiltonian Sign Decomposition)**.** *Under Assumptions 0.1 and 3.1, the quadratic Hamiltonian decomposes into:*

$$H^{(2)} = H^{(2)}_{\mathrm{pos}} + H^{(2)}_{\mathrm{bg}},$$

where $H^{(2)}_{\mathrm{pos}}$ contains the positive electric kinetic channels and the nonnegative magnetic square terms determined by the canonical pairing class, while $H^{(2)}_{\mathrm{bg}}$ contains the background-dependent terms. The sign of $H^{(2)}_{\mathrm{bg}}$ is background-dependent and is therefore not controlled in complete generality by the formal Hamiltonian structure alone.

Proof. The positive channels come from the kinetic terms already fixed in Sections 3 and 5 together with the magnetic square terms in the split Hamiltonian. Background-curvature cross terms are separated as the residual background contribution. The displayed decomposition in Appendix I records this separation explicitly. □

9.3 Classical Stability Regimes

Theorem 9.3 (Quadratic Hamiltonian Sign Classification)**.** *In the fixed positive-metric split regime, the quadratic Hamiltonian sign classification is as follows.*

(i) If the background contribution $H^{(2)}_{\mathrm{bg}}$ vanishes, then the quadratic Hamiltonian is nonnegative.

(ii) If $H^{(2)}_{\mathrm{bg}}$ does not vanish but is bounded below by the background class under consideration, then the quadratic Hamiltonian is bounded below and the corresponding regime is classically stable at quadratic level.

(iii) If the background contribution is not sign-controlled, then the formal Hamiltonian structure alone does not exclude indefinite channels, and the corresponding regime is only conditionally stable or potentially unstable.

Proof. The three regimes are exactly the cases distinguished by the sign behavior of $H^{(2)}_{\mathrm{bg}}$ in Proposition 9.2. □

10 Gravitational Restriction

10.1 The Codimension-One Sector

Section 10 concerns only the codimension-one specialization of the admissible compatibility theory. Thus N has rank one and

$$\mathfrak{so}(1) = 0.$$

The internal momentum-map mechanism of Sections 4–7 is therefore trivial in this sector.

10.2 Codimension-One Compatibility Identities

Proposition 10.1 (Codimension-One Contracted Identities)**.** *Under the codimension-one realization assumptions of this section, let γ denote the induced hypersurface metric, let K denote the*

realized second fundamental form, and let n^μ and $h^\nu{}_i$ denote the normal and tangential projectors. Then the contracted Gauss and Codazzi identities take the form

$$R(\gamma) + (\mathrm{tr}_\gamma K)^2 - |K|^2_\gamma = 2G_{\mu\nu}n^\mu n^\nu,$$

$$\nabla^j\big(K_{ij} - (\mathrm{tr}_\gamma K)\gamma_{ij}\big) = -\,G_{\mu\nu}n^\mu h^\nu{}_i.$$

Proof. [1] gives the contracted formulas in the form

$$2G_{\mu\nu}n^\mu n^\nu - R(\gamma) - (\mathrm{tr}_\gamma K)^2 + |K|^2_\gamma = 0,$$

$$G_{\mu\nu}n^\mu h^\nu{}_i + \nabla^j\big(K_{ij} - (\mathrm{tr}_\gamma K)\gamma_{ij}\big) = 0.$$

Rearranging yields the displayed identities. Appendix J records the codimension-one notation used here. □

Proposition 10.2 (Zero-Right-Hand-Side Specialization)**.** *Assume the codimension-one setting of Proposition 10.1 and, in addition, the ambient-Einstein projection restrictions*

$$G_{\mu\nu}n^\mu n^\nu = 0, \qquad G_{\mu\nu}n^\mu h^\nu{}_i = 0.$$

Then the contracted identities reduce to

$$R(\gamma) + (\mathrm{tr}_\gamma K)^2 - |K|^2_\gamma = 0, \qquad \nabla^j\big(K_{ij} - (\mathrm{tr}_\gamma K)\gamma_{ij}\big) = 0.$$

Proof. Substitute the stated vanishing Einstein projections into Proposition 10.1. □

10.3 Geometric Interpretation of the Mixed Variable

Proposition 10.3 (Conditional Identification with a Second Fundamental Form)**.** *Assume, in addition to Assumption 0.1, that there exists an embedded codimension-one hypersurface with induced metric γ, chosen unit normal field ν, and second fundamental form equal to the given symmetric tensor K. Then the rank-one mixed field is interpreted as the scalar-valued second fundamental form by*

$$K(X, Y) = K_{ij}\nu.$$

Under this realization hypothesis, the codimension-one formulas of Propositions 10.1 and 10.2 are read as standard contracted hypersurface formulas.

Proof. In rank one the mixed field takes values in a line bundle. After choosing a unit normal section ν, every realized second fundamental form is written in the displayed form. The interpretation requires the stated realization hypothesis; it is not automatic from rank one alone. Appendix J records the codimension-one translation. □

10.4 Restricted Formula Comparison

Theorem 10.4 (Codimension-One Formula Comparison)**.** *At formula level only, and only under the codimension-one realization hypothesis of Proposition 10.3 and the ambient-Einstein projection restrictions of Proposition 10.2, the contracted formulas of Proposition 10.2 reduce, within the fixed-metric internal-gauge regime, to the standard hypersurface Hamiltonian and momentum constraint equations.*

Proof. Proposition 10.3 supplies the geometric interpretation of the rank-one mixed field, and Proposition 10.2 supplies the zero-right-hand-side specialization. Under exactly those hypotheses, the displayed contracted formulas reduce to the standard hypersurface constraint equations. □

Proposition 10.5 (Restricted Codimension-One Scope)**.** *The codimension-one result remains a restricted hypersurface constraint comparison within the fixed-metric internal-gauge regime.*

Proof. Assumption 0.1 fixes the spacetime metric and excludes lapse-shift variation. In codimension one, $\mathfrak{so}(1) = 0$. Hence the codimension-one result remains restricted to hypersurface constraint identities and does not produce ADM phase space, a diffeomorphism constraint system, or the Dirac algebra. □

11 Structural Consequences

The paper determines the completed formal classical realization of the forced compatibility law within the fixed-metric internal-gauge regime. It passes from the covariant variational current and presymplectic structure to split canonical variables, Hamiltonian flow, the Gauss surface, the formal reduced observable algebra, closure, projected linearized symbol, reduced phase-space count, and quadratic Hamiltonian sign classification.

Only as a restricted codimension-one consequence does the realization hypothesis identifying the rank-one mixed field with a second fundamental form, together with the ambient-Einstein projection restrictions of Section 10, yield the formula-level comparison with the standard hypersurface constraint formulas.

The result remains confined to the fixed-metric internal-gauge regime and, in codimension one, to the restricted comparison of Theorem 10.4. Quantization lies beyond the present paper.

A First Variation and Variational Current

Starting from

$$L_{\mathrm{curv}} = \frac{1}{2} \left\langle \mathcal{F}^{\oplus}, \star\mathcal{F}^{\oplus} \right\rangle_{\mathrm{can}},$$

the curvature variation identity is

$$\delta\mathcal{F}^{\oplus} = D^{\oplus}(\delta\omega^{\oplus}).$$

Therefore

$$\delta L_{\mathrm{curv}} = \left\langle D^{\oplus}(\delta\omega^{\oplus}), \star\mathcal{F}^{\oplus} \right\rangle_{\mathrm{can}}.$$

Covariant integration by parts, with $(D^{\oplus})^{\dagger}$ the formal adjoint for the Hodge-paired density convention, gives

$$\delta L_{\mathrm{curv}} = \left\langle \delta\omega^{\oplus}, (D^{\oplus})^{\dagger}\mathcal{F}^{\oplus}, + \right\rangle_{\mathrm{can}} \mathrm{d} \left\langle \delta\omega^{\oplus}, \star\mathcal{F}^{\oplus} \right\rangle_{\mathrm{can}},$$

which proves Proposition 1.2.

For the admissible block variation

$$\delta\omega^{\oplus} = \begin{pmatrix} 0 & -\delta K^{*} \\ \delta K & \delta A \end{pmatrix}, \qquad \star\mathcal{F}^{\oplus} = \begin{pmatrix} \star\mathcal{G} & -(\star\mathcal{C})^{*} \\ \star\mathcal{C} & \star\mathcal{S} \end{pmatrix},$$

the pairing is

$$\left\langle \delta\omega^{\oplus}, \star\mathcal{F}^{\oplus} \right\rangle_{\mathrm{can}} = -\mathrm{tr}(\delta\omega^{\oplus} \star \mathcal{F}^{\oplus}).$$

Multiplying the blocks yields

$$\delta\omega^{\oplus} \star \mathcal{F}^{\oplus} = \begin{pmatrix} -\delta K^* \star \mathcal{C} & -\delta K^* \star \mathcal{S} \\ \delta K \star \mathcal{G} + \delta A \star \mathcal{C} & -\delta K(\star\mathcal{C})^* + \delta A \star \mathcal{S} \end{pmatrix}.$$

Taking the trace gives

$$\mathrm{tr}(\delta\omega^{\oplus} \star \mathcal{F}^{\oplus}) = -\mathrm{tr}(\delta K^* \star \mathcal{C}) - \mathrm{tr}(\delta K(\star\mathcal{C})^*) + \mathrm{tr}(\delta A \star \mathcal{S}).$$

Since the two mixed terms are equal by block adjunction in the canonical trace pairing,

$$\Theta_L(\delta\omega^{\oplus}) = -\langle \delta A, \star\mathcal{S}\rangle_{\mathrm{can}} + 2\langle \delta K, \star\mathcal{C}\rangle_{\mathrm{can}}.$$

B Presymplectic Current and Hypersurface Form

By definition,

$$\omega_L(\delta_1, \delta_2) = \delta_1\Theta_L(\delta_2) - \delta_2\Theta_L(\delta_1).$$

Substituting the block formula for Θ_L gives

$$\omega_L(\delta_1, \delta_2) = -\delta_1 \langle \delta_2 A, \star\mathcal{S}\rangle_{\mathrm{can}} + \delta_2 \langle \delta_1 A, \star\mathcal{S}\rangle_{\mathrm{can}} + 2\delta_1 \langle \delta_2 K, \star\mathcal{C}\rangle_{\mathrm{can}} - 2\delta_2 \langle \delta_1 K, \star\mathcal{C}\rangle_{\mathrm{can}}.$$

The hypersurface presymplectic form is

$$\Omega_{\Sigma_t}(\delta_1, \delta_2) = \int_{\Sigma_t} \omega_L(\delta_1, \delta_2).$$

If $\delta_\epsilon\omega^{\oplus} = D^{\oplus}\epsilon$, then the Noether identity yields

$$\iota_{\delta_\epsilon}\omega_L = \mathrm{d}\mathcal{B}_\epsilon$$

for a boundary current $\mathcal{B}_\epsilon$. Therefore

$$\iota_{\delta_\epsilon}\Omega_{\Sigma_t} = \int_{\Sigma_t} \mathrm{d}\mathcal{B}_\epsilon$$

vanishes under Assumption 0.1(v).

To identify Ω_{Σ_t} with the canonical symplectic form, use the split conventions of Assumption 3.1. Identify $\Sigma_t \cong \Sigma$ by the fixed product decomposition, and decompose the relevant curvature-star blocks as

$$\star\mathcal{S} = *_\Sigma E_\perp + dt \wedge *_\Sigma B_\perp, \qquad \star\mathcal{C} = *_\Sigma E_K + dt \wedge *_\Sigma B_K.$$

The hypersurface pullback annihilates the dt-term and retains the spatial Hodge-dual electric coefficient:

$$\iota_t^*(\star\mathcal{S}) = *_\Sigma E_\perp, \qquad \iota_t^*(\star\mathcal{C}) = *_\Sigma E_K.$$

Therefore

$$\Omega_{\Sigma_t} = \int_\Sigma \left(\delta E_\perp \wedge \delta A + 2\delta E_K \wedge \delta K\right),$$

and substituting the Legendre relations

$$p_A = E_\perp, \qquad p_K = 2E_K$$

gives

$$\Omega_{\Sigma_t} = \int_\Sigma \left(\delta p_A \wedge \delta A + \delta p_K \wedge \delta K\right),$$

which is the canonical symplectic form of Proposition 3.6.

C Legendre Transform and Canonical Hamiltonian

Under the fixed product split of Assumption 3.1, write the electric channel as $E^{\oplus} = (E_{\perp}, E_K)$. The quadratic curvature density induces the split Lagrangian functional

$$L = \int_{\Sigma} \frac{1}{2} \left(E_{\perp}^2 + 2E_K^2 \right) \operatorname{vol}_{\gamma} - V_{\mathrm{mag}},$$

with

$$V_{\mathrm{mag}} = \frac{1}{2} \int_{\Sigma} \left\langle B^{\oplus}, B^{\oplus} \right\rangle_{\mathrm{can}} \operatorname{vol}_{\gamma}.$$

The kinetic block is diagonal in $(E_{\perp}, E_K)$, so the pointwise Legendre inversion is immediate. Differentiate L to obtain

$$p_A = \frac{\partial L}{\partial E_{\perp}} = E_{\perp}, \qquad p_K = \frac{\partial L}{\partial E_K} = 2E_K.$$

Here $\mathcal{H}$ denotes the Hamiltonian density and H the Hamiltonian functional. Hence

$$E_{\perp} = p_A, \qquad E_K = \frac{1}{2} p_K.$$

The Legendre transform gives

$$\begin{aligned} H &= p_A E_{\perp} + p_K E_K - L \\ &= \int_{\Sigma} \left(p_A^2 + \frac{1}{2} p_K^2 - \frac{1}{2} \left(p_A^2 + 2 \left(\frac{1}{2} p_K \right)^2 \right) \right) \operatorname{vol}_{\gamma} + V_{\mathrm{mag}} \\ &= \int_{\Sigma} \left(\frac{1}{2} p_A^2 + \frac{1}{4} p_K^2 + \frac{1}{2} \left\langle B^{\oplus}, B^{\oplus} \right\rangle_{\mathrm{can}} \right) \operatorname{vol}_{\gamma}. \end{aligned}$$

Hence the Hamiltonian density is

$$\mathcal{H} = \frac{1}{2} p_A^2 + \frac{1}{4} p_K^2 + \frac{1}{2} \left\langle B^{\oplus}, B^{\oplus} \right\rangle_{\mathrm{can}},$$

and the Hamiltonian functional is

$$H = \int_{\Sigma} \mathcal{H} \operatorname{vol}_{\gamma}.$$

The formal canonical one-form and symplectic form are therefore

$$\Theta = \int_{\Sigma} (p_A \, \delta A + p_K \, \delta K), \qquad \Omega = -\delta \Theta.$$

This is the formal canonical symplectic form of Sections 4–6.

D Momentum Map and Gauss Constraint

Let $\epsilon \in \mathfrak{g}_N$. The infinitesimal gauge vector field is

$$X_{\epsilon} = (D_A \epsilon, \epsilon \cdot K, [\epsilon, p_A], \epsilon \cdot p_K).$$

Contracting with Ω gives

$$\iota_{X_{\epsilon}} \Omega = \int_{\Sigma} \left(\delta p_A \, D_A \epsilon - \delta A \, [\epsilon, p_A] + \delta p_K \, (\epsilon \cdot K) - \delta K \, (\epsilon \cdot p_K) \right).$$

Integrating by parts and regrouping yields

$$\iota_{X_\epsilon}\Omega = \mathrm{d}\left(\int_\Sigma \langle \epsilon, D_A p_A + p_K K^* - K p_K^* \rangle_{\mathrm{can}}\ \mathrm{vol}_\gamma\right),$$

which proves Proposition 4.1. The formal tangent equation to $\mathcal{S}_{\mathrm{G}}$ is obtained by linearizing the integrated Gauss functional with respect to the same pairing. This is the momentum-map identity invoked in Proposition 4.1.

E Hamiltonian Closure and Reduced Observable Algebra

The Hamilton equations are

$$\dot{A} = \frac{\delta H}{\delta p_A} = p_A, \qquad \dot{K} = \frac{\delta H}{\delta p_K} = \frac{1}{2} p_K,$$

$$\dot{p}_A = -\frac{\delta H}{\delta A}, \qquad \dot{p}_K = -\frac{\delta H}{\delta K}.$$

Using $p_A = E_\perp$ and $p_K = 2E_K$ converts these into the second-order Euler system determined by the split Lagrangian.

Write

$$\star\mathcal{F}^\oplus = \mathcal{E}_\star^\oplus + dt \wedge \mathcal{B}_\star^\oplus,$$

where $\mathcal{E}_\star^\oplus = *_\Sigma \mathcal{E}^\oplus$, and write the admissible connection as $\omega^\oplus = \omega_0^\oplus\, dt + \omega_\Sigma^\oplus$. Then

$$D^\oplus \star \mathcal{F}^\oplus = dt \wedge \left(\partial_t \mathcal{E}_\star^\oplus + [\omega_0^\oplus, \mathcal{E}_\star^\oplus] - D_{\omega_\Sigma}\mathcal{B}_\star^\oplus\right) + D_{\omega_\Sigma}\mathcal{E}_\star^\oplus.$$

Hence $D^\oplus \star \mathcal{F}^\oplus = 0$ is equivalent to

$$D_{\omega_\Sigma}\mathcal{E}_\star^\oplus = 0, \qquad \partial_t \mathcal{E}_\star^\oplus + [\omega_0^\oplus, \mathcal{E}_\star^\oplus] - D_{\omega_\Sigma}\mathcal{B}_\star^\oplus = 0.$$

The first equation is the Gauss channel; the second is the evolution channel.

Projecting the Gauss channel to the normal block gives

$$D_A p_A + p_K K^* - K p_K^* = 0,$$

which is Theorem 4.2. Projecting the evolution channel to the normal and mixed admissible blocks yields the Euler equations obtained by varying the split Lagrangian with respect to A and K. Hence the Hamilton equations agree with the projected evolution channel after the Legendre substitution.

For the Gauss generators,

$$\{\langle J, \epsilon_1\rangle, \langle J, \epsilon_2\rangle\} = \langle J, [\epsilon_1, \epsilon_2]\rangle$$

because the Hamiltonian vector field of $\langle J, \epsilon\rangle$ is the infinitesimal internal gauge action, and the commutator of those infinitesimal actions is the bracket $[\epsilon_1, \epsilon_2]$. Since H is built from gauge-covariant electric and magnetic pairings,

$$\{H, \langle J, \epsilon\rangle\} = 0.$$

Finally, if F and G are gauge invariant, then

$$\{\{F, G\}, \langle J, \epsilon\rangle\} = \{\{F, \langle J, \epsilon\rangle\}, G\} + \{F, \{G, \langle J, \epsilon\rangle\}\} = 0$$

by the Jacobi identity. Hence gauge-invariant observables are closed under the canonical bracket.

F Closure and Constraint-Propagation Calculations

Since

$$\{\langle J,\epsilon_1\rangle,\langle J,\epsilon_2\rangle\} = \langle J,[\epsilon_1,\epsilon_2]\rangle, \qquad \{H,\langle J,\epsilon\rangle\} = 0,$$

the internal Gauss generators form a closed first-class subsystem and the Hamiltonian preserves the Gauss surface.

If $(A(t),K(t),p_A(t),p_K(t))$ is a Hamiltonian trajectory, then for every $\epsilon \in \mathfrak{g}_N$,

$$\frac{d}{dt}\langle J,\epsilon\rangle = \{\langle J,\epsilon\rangle, H\} = 0.$$

Hence $\langle J,\epsilon\rangle$ is constant along the flow. If it vanishes at one time, it vanishes throughout the interval of formal Hamiltonian evolution.

To compare canonical and covariant closure, use the projected decomposition

$$D^{\oplus} \star \mathcal{F}^{\oplus} = dt \wedge \big(\partial_t \mathcal{E}^{\oplus}_\star + [\omega_0^{\oplus}, \mathcal{E}^{\oplus}_\star] - D_{\omega_\Sigma}\mathcal{B}^{\oplus}_\star\big) + D_{\omega_\Sigma}\mathcal{E}^{\oplus}_\star.$$

The temporal channel

$$D_{\omega_\Sigma}\mathcal{E}^{\oplus}_\star = 0$$

is the Gauss channel after the Legendre substitution, while the spatial channel

$$\partial_t \mathcal{E}^{\oplus}_\star + [\omega_0^{\oplus}, \mathcal{E}^{\oplus}_\star] - D_{\omega_\Sigma}\mathcal{B}^{\oplus}_\star = 0$$

is the projected evolution channel. Thus the projected covariant identity structure and the canonical closure and propagation statements agree in the fixed split regime.

G Linearization and Projected Principal Symbol

Fix an admissible background $(A^{(0)},K^{(0)},p_A^{(0)},p_K^{(0)})$ satisfying the classical compatibility system. Write perturbations

$$a := A - A^{(0)}, \qquad \kappa := K - K^{(0)}, \qquad \pi_A := p_A - p_A^{(0)}, \qquad \pi_K := p_K - p_K^{(0)}.$$

Linearizing the projected split system gives

$$\delta\big(D_{\omega_\Sigma}\mathcal{E}^{\oplus}_\star\big) = 0, \qquad \delta\big(\partial_t \mathcal{E}^{\oplus}_\star + [\omega_0^{\oplus}, \mathcal{E}^{\oplus}_\star] - D_{\omega_\Sigma}\mathcal{B}^{\oplus}_\star\big) = 0.$$

Replacing ∂_t and ∂_i by the dual variables (τ,ξ_i) yields the projected principal-symbol model:

$$\sigma_{\mathrm{proj}}(\tau,\xi)\begin{pmatrix} a_{\mathrm{proj}} \\ \kappa_{\mathrm{proj}} \end{pmatrix} = \begin{pmatrix} \tau^2 a_{\mathrm{proj}} + Q_A(\xi)\, a_{\mathrm{proj}} + C_A(\xi,K^{(0)})\,\kappa_{\mathrm{proj}} \\ \tau^2 \kappa_{\mathrm{proj}} + Q_K(\xi)\,\kappa_{\mathrm{proj}} + C_K(\xi,K^{(0)})\, a_{\mathrm{proj}} \end{pmatrix},$$

where $Q_A(\xi)$ and $Q_K(\xi)$ are the projected spatial symbol blocks and C_A, C_K are the background-coupling symbol terms determined by the admissible split. At generic admissible backgrounds, this symbol is quadratic in (τ,ξ) on the projected normal block a_{proj} and mixed block κ_{proj}. At the special flat compatible backgrounds with $K^{(0)} = 0$, the projected mixed-block kernel direction inherited from the reduced-symbol degeneracy of [1] lies in the kernel of the projected principal symbol. More explicitly,

$$\sigma_{\mathrm{proj}}(\tau,\xi)\begin{pmatrix} 0 \\ \kappa_{\mathrm{deg}}(\xi) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Thus $\kappa_{\mathrm{deg}}(\xi)$ is a mixed-block kernel direction of the projected principal symbol.

H Degree-of-Freedom Counting

The spatial normal-connection block A_Σ contributes

$$n\frac{k(k-1)}{2}$$

configuration variables per spatial point, where $n = \dim \Sigma$ and $k = \mathrm{rank}(N)$. The spatial mixed block K_Σ contributes

$$\frac{n(n+1)}{2}k$$

configuration variables per spatial point.

Therefore the unreduced canonical phase space has

$$2\left(n\frac{k(k-1)}{2} + \frac{n(n+1)}{2}k\right)$$

components per spatial point. The internal gauge algebra has dimension

$$\dim \mathfrak{so}(N) = \frac{k(k-1)}{2},$$

and the Gauss generators are first class. Hence internal gauge reduction removes

$$2\dim \mathfrak{so}(N) = k(k-1)$$

phase-space components, leaving

$$2\left(n\frac{k(k-1)}{2} + \frac{n(n+1)}{2}k\right) - k(k-1)$$

reduced classical phase-space components per spatial point.

I Quadratic Hamiltonian and Stability Regimes

Let $H^{(2)}$ denote the second-order part of the split Hamiltonian about a fixed admissible background. In the fixed positive-metric split regime it has the form

$$H^{(2)} = H^{(2)}_{\mathrm{pos}} + H^{(2)}_{\mathrm{bg}},$$

where $H^{(2)}_{\mathrm{pos}}$ contains the positive electric kinetic terms

$$\frac{1}{2}\|\pi_A\|^2 + \frac{1}{4}\|\pi_K\|^2$$

and the nonnegative magnetic square terms, while $H^{(2)}_{\mathrm{bg}}$ collects the background-dependent residual terms.

$$H^{(2)} = \left(\frac{1}{2}\|\pi_A\|^2 + \frac{1}{4}\|\pi_K\|^2 + \frac{1}{2}\|B^{(1)}\|^2\right) + H^{(2)}_{\mathrm{bg}},$$

so the positive electric and magnetic-square channels are displayed separately from the background-dependent remainder.

No universal closed-form sign classification of $H^{(2)}_{\mathrm{bg}}$ is asserted. If $H^{(2)}_{\mathrm{bg}} = 0$, then $H^{(2)} \geq 0$. If $H^{(2)}_{\mathrm{bg}}$ is bounded below in the background class under consideration, then $H^{(2)}$ is bounded below. If $H^{(2)}_{\mathrm{bg}}$ is not sign-controlled, the formal Hamiltonian structure alone does not rule out indefinite channels. This decomposition is the sign input used in Proposition 9.2 and Theorem 9.3.

J Restricted-Sector Calculations

In codimension one, and under the realization assumptions of Section 10, N has rank one, so

$$\mathfrak{so}(N) = \mathfrak{so}(1) = 0.$$

The internal momentum-map and Gauss-generator mechanism of Sections 4–5 is therefore trivial in codimension one.

Assume the codimension-one data arise from an embedded hypersurface with chosen unit normal field ν in an ambient metric, together with the realization hypotheses needed to identify the given symmetric tensor K with the second fundamental form of that hypersurface. Then the mixed variable is written as

$$K(X, Y) = K_{ij}\nu,$$

so the present compatibility field is interpreted as the scalar-valued second fundamental form K_{ij}.

[1] gives the contracted codimension-one identities in the form

$$R(\gamma) + (\mathrm{tr}_\gamma K)^2 - |K|^2_\gamma = 2G_{\mu\nu}n^\mu n^\nu,$$

$$\nabla^j\big(K_{ij} - (\mathrm{tr}_\gamma K)\gamma_{ij}\big) = -\,G_{\mu\nu}n^\mu h^\nu{}_i.$$

Under the additional ambient-Einstein projection restrictions

$$G_{\mu\nu}n^\mu n^\nu = 0, \qquad G_{\mu\nu}n^\mu h^\nu{}_i = 0,$$

these reduce to the zero-right-hand-side formulas yielding the standard hypersurface Hamiltonian and momentum constraint equations.

These identities do not imply full ADM equivalence. The present theory keeps the spacetime metric fixed, does not vary lapse or shift, and does not quotient diffeomorphisms. Because $\mathfrak{so}(1) = 0$, the internal gauge mechanism in codimension one is not replaced by the ADM diffeomorphism-constraint system or Dirac algebra. The codimension-one statement is therefore restricted to the comparison under the stated realization and ambient-curvature hypotheses.

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